

```
# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# load dataset
df = pd.read_csv('/content/TeachingRatings.csv')
print(df.head(4))
print(df.tail(4))
```

```
   rownames minority  age  gender credits  beauty  eval division native \
0         1      yes   36  female  more  0.289916  4.3    upper    yes
1         2      no   59   male  more -0.737732  4.5    upper    yes
2         3      no   51   male  more -0.571984  3.7    upper    yes
3         4      no   40  female  more -0.677963  4.3    upper    yes

   tenure  students  allstudents  prof
0    yes         24           43     1
1    yes         17           20     2
2    yes         55           55     3
3    yes         40           46     4

   rownames minority  age  gender credits  beauty  eval division native \
459       460      no   32   male  more  1.231394  4.3    upper    yes
460       461      yes  42  female  more  0.420400  3.3    upper    no
461       462      yes  42  female  more  0.420400  3.2    upper    no
462       463      yes  42  female  single 0.420400  4.1    lower    no

   tenure  students  allstudents  prof
459    yes         52           86    93
460    yes         52           67    94
461    yes         54           66    94
462    yes         28           35    94
```

Question 1

Identify all duplicate cases using prof. Using all observations, find the average and standard deviation for age. Repeat the analysis by first filtering the dataset to include one observation for each instructor with a total number of observations restricted to 94.

```
# Identify duplicate cases using prof
print(df.duplicated(subset=['prof']).sum())
```

369

There are 369 duplicate instructor observations because the dataset has 463 rows but only 94 unique instructors.

```
# Average and standard deviation of age using all observations
```

```
# Mean age  
mean_age = df['age'].mean()
```

```
# Standard deviation of age  
std_age = df['age'].std()
```

```
print("Average age:", mean_age)  
print("Standard deviation:", std_age)
```

```
Average age: 48.365010799136066  
Standard deviation: 9.802742037864817
```

```
# Keep only one observation for each instructor  
df_unique = df.drop_duplicates(subset=['prof'])
```

```
print(df_unique.shape)
```

```
(94, 13)
```

```
# Average and standard deviation of age using 94 observations.
```

```
mean_age_unique = df_unique['age'].mean()  
std_age_unique = df_unique['age'].std()
```

```
print("Average age:", mean_age_unique)  
print("Standard deviation:", std_age_unique)
```

```
Average age: 47.5531914893617  
Standard deviation: 10.256513295154951
```

Question 2

Using a bar chart, demonstrate if instructors teaching lower-division courses receive higher average teaching evaluations.

```
# Calculate average evaluation by division  
avg_eval = df.groupby('division')['eval'].mean()
```

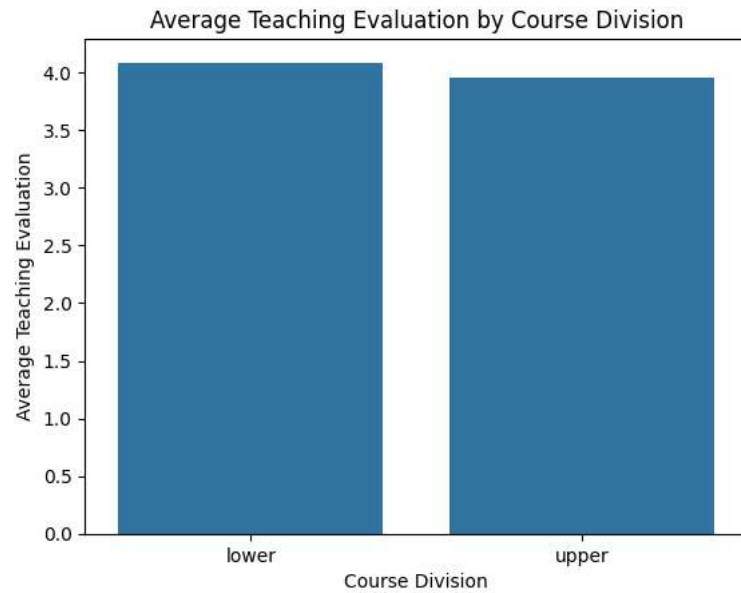
```
print(avg_eval)
```

```
# Bar chart  
sns.barplot(x=avg_eval.index, y=avg_eval.values)
```

```
plt.title('Average Teaching Evaluation by Course Division')  
plt.xlabel('Course Division')  
plt.ylabel('Average Teaching Evaluation')
```

```
plt.show()
```

```
division  
lower    4.087261  
upper    3.952614  
Name: eval, dtype: float64
```



Conclusion

Yes. Lower-division courses have a higher average teaching evaluation than upper-division courses:

1. Lower division: 4.087
2. Upper division: 3.953

The difference is approximately 0.135 points.

So the bar chart provides evidence that instructors teaching lower-division courses received higher average evaluations in this dataset.

Question 3

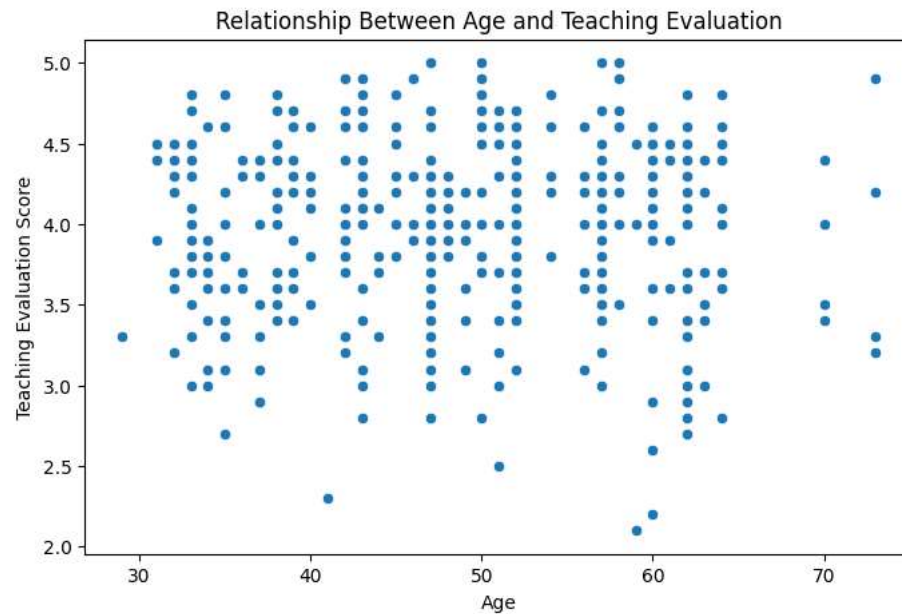
Plot the relationship between age and teaching evaluation scores.

```
plt.figure(figsize=(8, 5))

sns.scatterplot(data=df, x='age', y='eval')

plt.title('Relationship Between Age and Teaching Evaluation')
plt.xlabel('Age')
plt.ylabel('Teaching Evaluation Score')

plt.show()
```

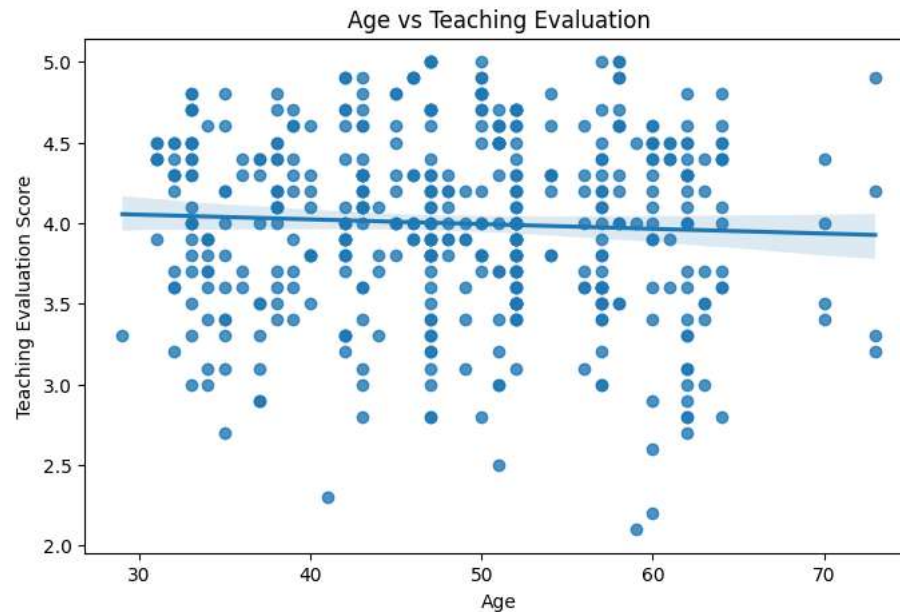


```
plt.figure(figsize=(8, 5))

sns.regplot(data=df, x='age', y='eval')

plt.title('Age vs Teaching Evaluation')
plt.xlabel('Age')
plt.ylabel('Teaching Evaluation Score')

plt.show()
```



Interpretation

The relationship appears to be weakly negative overall: as age increases, teaching evaluation tends to decrease slightly.

However, you should not say that age causes lower evaluations. A scatterplot shows association, not causation.

✓ Question 4

Using gender-differentiated scatter plots, plot the relationship between age and teaching evaluation scores.

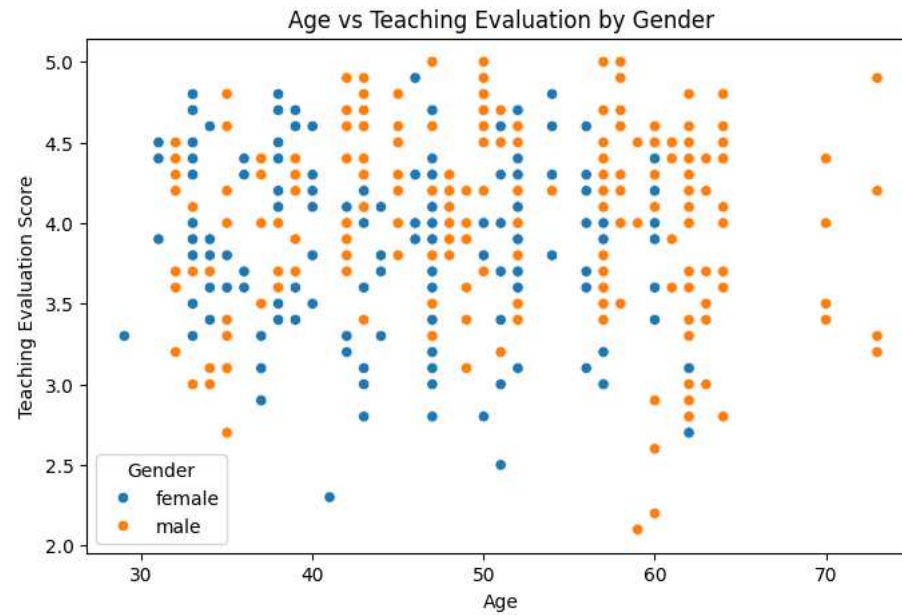
```
plt.figure(figsize=(8, 5))

sns.scatterplot(
    data=df,
    x='age',
    y='eval',
    hue='gender'
)

plt.title('Age vs Teaching Evaluation by Gender')
```

```
plt.xlabel('Age')
plt.ylabel('Teaching Evaluation Score')
plt.legend(title='Gender')

plt.show()
```



```
sns.lmplot(
    data=df,
    x='age',
    y='eval',
    col='gender',
    height=5,
    aspect=1
)

plt.show()
```

